

Mateus Appuwan Situmorang

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Informatics graduate from Sanata Dharma University with a strong foundation in full-stack development and a developing focus on cybersecurity engineering. Experienced in delivering secure, scalable, and well-documented digital solutions, supported by a security-focused internship at *Badan Siber dan Sandi Negara* (BSSN) and multiple end-to-end projects. Adept at bridging technical execution with team leadership and cross-functional communication.

WORK EXPERIENCE

Full-stack Developer, System Engineer Intern - *Badan Siber dan Sandi Negara* (November 2025 – May 2026)

South Jakarta, DKI Jakarta

- Led two concurrent security projects a malware analysis sandbox and a secure Android key-entry app coordinating a 3-person dev team across planning, development, and system integration.
- Built an automated malware sandbox integrating Nginx, Docker, Python, MISP, VirusTotal, and Arkime; accurately detected WannaCry with a malscore of 10/10 while scoring benign files at 0–1/10.
- Developed a Flutter/Dart Android app with secure login, user profile management, and cryptographic key-entry compliant with EAL 3 standards; all user data encrypted and user-key-protected.
- Defined acceptance criteria aligned with Common Criteria and applied security-by-design across the SDLC covering access control, data flow hardening, and application hardening.
- Managed team workflow via GitHub (branching, PRs, code reviews) and documented system architecture in Miro.

PROJECT EXPERIENCE

Full-stack Developer - *YogTrip - Yogyakarta Travel Package Recommendation System* (August 2024 – July 2025)

Final Thesis Project – Sleman, Special Region of Yogyakarta

- Designed and implemented a hybrid recommendation engine leveraging Knowledge Graph structures and the Path Ranking Algorithm (PRA), integrating MySQL and Neo4j to enable advanced relational graph data mapping for personalized travel recommendations.
- Achieved strong evaluation benchmarks validating recommendation quality and system reliability: Precision@k 0.72, Recall@k 0.84, NDCG@k 0.75–0.82, and Hit Rate@k up to 0.97.

Web Development - *LOCALINE - Web System Development (Union in Technology)* (September 2024 – October 2024)

IT Days 2024 – Sleman, Special Region of Yogyakarta

- Collaborated within a cross-functional team to build a React-based web platform designed to support Indonesian SME (UMKM) digitalization and expand local market access.
- Delivered a functional, competition-ready platform that secured 3rd place at IT Days 2024, demonstrating both technical quality and practical readiness for real-world SME adoption.

Full-stack Developer - *Mavis Betta Fish - Web-based E-commerce System Development* (August 2023 – December 2023)

Information Technology Project – Sleman, Special Region of Yogyakarta

- Engineered an end-to-end e-commerce workflow encompassing front-end development, back-end API design, and relational database architecture to support product catalog management and transaction processing.
- Delivered a reliable, secure shopping experience with zero data integrity incidents, ensuring consistent product and transaction data handling across all user interactions.

EDUCATION

Bachelor of Computer Science (August 2021 – July 2025)

Sanata Dharma University – Special Region of Yogyakarta, Indonesia

- GPA 3.39/4.00
- Thesis Title: “Yogyakarta Tour Package Recommendation System Based on Knowledge Graph with Path Ranking Algorithm (PRA)”

Public Senior High School 02 Dente Teladas (July 2019 – May 2021)

Dente Teladas, Tulang Bawang, Lampung

SKILLS

- **Software Engineering:** Flutter/Dart, JavaScript (Node.js, React), Python, PHP, HTML/CSS.
- **Data & Databases:** MySQL, PostgreSQL, Neo4j, Knowledge Graphs, Evaluation Metrics (Precision / Recall / NDCG / Hit Rate).
- **Security Engineering (High-level):** Secure SDLC, EAL 3 Compliance, Application Hardening, Access Control, Data Encryption, Malware Analysis (Docker, MISP, VirusTotal, MITRE ATT&CK)
- **Infrastructure & Tools:** Linux / Ubuntu Server, Docker, GitHub (version control & code review), Postman (API testing), Miro (system documentation).
- **Languages:** Indonesian (Native), English (Intermediate).

CERTIFICATIONS

- Essential Skills: Digital Disruption & Transformation** - GNIK (*Gerakan Nasional Indonesia Kompeten*) (January 2025)
- Essential Skills: Design Thinking** - GNIK (*Gerakan Nasional Indonesia Kompeten*) (December 2025)
- Python (Basic)** - HackerRank (October 2025)
- SAP01 - SAP Overview** - SAP Course (June 2025)
- Programming for Everybody (Getting Started with Python)** - University of Michigan (July 2025)
- Ultimate CSSLP Masterclass - Secure Software Development** - Udemy (July 2025)

ORGANIZATIONAL & VOLUNTEER EXPERIENCE

- Media Communication and Information (BEM)** (December 2022 – December 2023)
Student Executive Board, Faculty of Science and Technology
- Managed the creation of all visual content (Instagram feeds, videos, photos) and coordinated documentation coverage for six major faculty work programs in close collaboration with the BEM Governor.
 - Maintained consistent public communication and a cohesive visual identity for the Faculty of Science and Technology across all published materials.
- Publication, Decoration & Documentation** (June 2024 – July 2024)
Community Service Program (KKN) at Sanata Dharma University 68, Slawu Gunungkidul Regency, Special Region of Yogyakarta
- Led rebranding efforts for local SME businesses in Slawu Village, organized an SME competition, and delivered individual education sessions on leveraging social media as a product promotion channel.
 - Produced the official event after-movie and managed all Instagram documentation, generating measurable growth in community enthusiasm and public interest in local SME initiatives.
- SIESEN (Vocalist)** (March 2023 – September 2023)
Sanata Dharma Initiation MMXXIII Sleman, Special Region of Yogyakarta
- Collaborated intensively with the creative team to compose original music, arrange melodies, and write lyrics for the university's new student orientation event, serving as lead vocalist during the live performance.
 - Performed original compositions before thousands of attendees, demonstrating strong stage presence, effective team collaboration, and confident public communication.